

7

Motion



Skateboard is one of the best applications of motion and its laws. When you hit a sidewalk, rock, or anything else that causes the skateboard to halt unexpectedly, you fly forward away from the board. As a result, unless acted upon by an imbalanced force, a skateboard at rest will remain at rest. A skateboard in motion will just keep rolling in the absence of friction or any other balancing force!



What's Inside

Topic(s)

- ✓ Topic-wise Notes
- ✓ Related NCERT Qs
- ✓ Caution Points
- ✓ Case-based Examples

Questions

- ✓ New Pattern Qs
- ✓ All Objective + Subjective Qs
- ✓ Diksha + Exemplar Qs
- ✓ Detailed Explanation

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TOPIC 1

MOTION

Learning Objectives

- Students will learn in detail about the types of motion.
- Students will learn about the reference point and reference frame.

Learning Outcomes

- Students will be able to explain the different types of motion.
- Students will be able to differentiate between the reference point and the reference frame.

Real Life Application

Do you remember? In the scorching summer days, when we come back to our homes from outside, the very first thing we do is switch on a ceiling fan or a table fan which delivers an amount of cool air at low temperatures which makes us feel relaxed.

You can easily observe that a fan is comprised of a motor connected to blades (usually three to four). These blades rotate at a speed to create a fast flow of air causing it to cool. How do we receive air from the fan? It is because the fan is in motion and motion is a change in the position of a body with respect to time.

Apart from this, in our daily life, we come across a number of objects that are said to be in motion. E.g., a flying aeroplane, a moving car, the needles of a clock etc. In this topic, let us learn more about motion and its types.



Types of Motion

There are five types of motion, we have discussed below:



- (1) **Linear Motion:** The motion of an object along the straight line is linear motion.
For example, a bullet shot from a pistol, swimming in a straight line.
- (2) **Circular Motion:** When an object is constantly moving in a circular path, it is called circular motion.
For example, the swirling of stone tied to a string, the hands of a clock and a satellite orbiting around a planet.
- (3) **Oscillatory Motion:** The to and fro motion of an object from its mean position is known as oscillatory motion.
Example: Pendulum Clock, tuning fork, the vibration of strings of musical instruments.
- (4) **Rotatory Motion:** Rotatory motion occurs when a body moves around a fixed axis without changing the radius of its motion.
For example, the motion of the wheel about its axis, steering wheel, spinning top, rotating fan and the motion of the earth about its axis around the sun.
- (5) **Translational Motion:** Translational motion is defined as the movement of a body from one point in space to another. Based on the type of path taken by an object, it can be rectilinear or curvilinear.
For example, a train moving on a track or a roller coaster ride.

Reference Point and Reference Frame

If you tell your friend that your house is 3 m away from the railway station. It might confuse him because you are not telling him your position in a precise way. Therefore, it requires a reference point as well as coordinates like he should move to left, right or in a particular direction.

Hence, it is described as the frame of reference.

A frame of reference is a combination of a reference point and a set of directions.

For example: When you tell your friend that your house direction is 3 m left to the railway station and you gave him set of instructions.

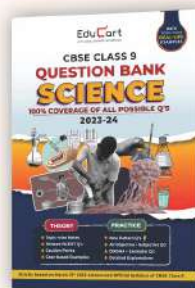
The particular point about which the position of an object is defined is called the reference point or origin.





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OBJECTIVE Type Questions

Multiple Choice Questions

[1 mark each]

1. For a moving object, the numerical ratio of displacement to distance is:

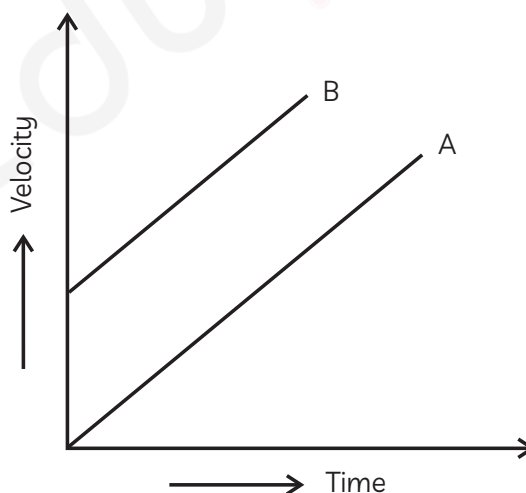
- (a) always less than 1
- (b) always equal to 1
- (c) always greater than 1
- (d) equal or less than 1

[NCERT Exemplar]

Ans. (c) equal or less than 1

Explanation: Displacement is the shortest distance between the initial and final position whereas distance is the actual length of the path travelled by an object. For a moving object, distance is always more than or equal to displacement. Thus, the ratio of displacement to distance is equal to or less than 1.

2. Two slopes in velocity-time graphs are shown in the given figure. Which slope indicates a higher rate of change?



Velocity-time graph

- (a) Only A
- (b) Only B
- (c) Both A and B
- (d) Can't be determined

Ans. (c) Both A and B

Explanation: Rate of change means acceleration. Here, acceleration is the same for both A and B because the two graphs are parallel lines and their slopes are equal.

Assertion-Reason

[A-R]

[1 mark each]

In the following questions, two statements in the form of an Assertion (A) and a Reason (R) have been put forward. Read both statements carefully and choose the most appropriate option:

- (a) Both (A) and (R) are true and (R) is the correct explanation of (A).
- (b) Both (A) and (R) are true but (R) is not the correct explanation of (A).
- (c) (A) is true but (R) is false.
- (d) (A) is false but (R) are true.

3. Assertion (A): Lion accelerated from rest at the rate of 6 m/s^2 .

Reason (R): The velocity attained by a lion in 3 sec is 2 m/s.

Ans. (a) Both A and R are true and R is not the correct explanation of A.

Explanation: Given,

Acceleration, $a = 6 \text{ m/s}^2$.

Time, $t = 3 \text{ sec}$

Initial velocity, $u = 0 \text{ m/s}$

$$a = \frac{v - u}{t}$$

$$6 = \frac{v - 0}{3}$$

$$v = 18 \text{ m/s}$$

Now, in reason, it's given

Final velocity, $v = 2 \text{ m/s}$

Time, $t = 3 \text{ sec}$

$$a = \frac{2 - 0}{3}$$

$$a = 0.66 \text{ m/s}^2$$

Fill in the Blanks

[1 mark each]

4. A coin is tossed and it starts falling freely. The displacement of the coin for a short interval is positive during and negative during if upward is considered a positive direction.

Ans. time of ascent; time of descent

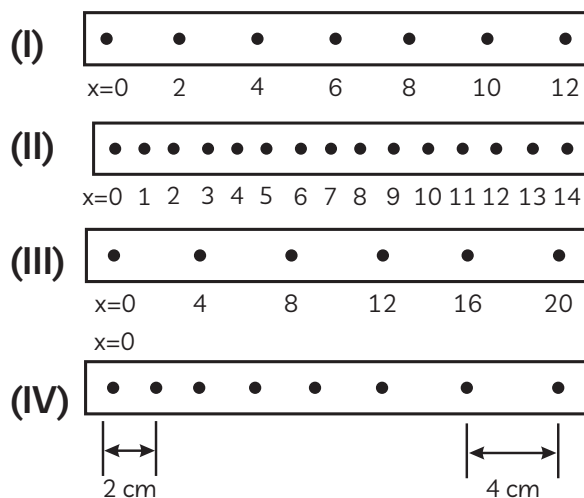
Explanation: Since the coin is rising during the ascent, its position is positive. Even though the velocity is reducing, it is now moving upward, in a positive direction. The coin is travelling under the influence of gravitational acceleration, which is always negative and acts vertically downwards.

CASE BASED Questions (CBQs)

[5 marks each]

Read the following passages and answer the questions that follow:

5. Imagine a car with a leaky engine that drips oil at a regular rate of 50 drops per second. As the car travels through town, it would leave a trace of oil on the street that would reveal information about the motion of the car. The trace of the oil in four different situations is shown. (The positions are given in centimetres.)



(A) Identify the type of motion by the car in situations I, II, III and IV.

	I	II	III	IV
(a)	Uniform	Non-uniform	Uniform	Non-uniform
(b)	Uniform	Uniform	Uniform	Non-uniform
(c)	Uniform	Uniform	Uniform	Non-uniform
(d)	Non-Uniform	Non-uniform	Non-uniform	Non-uniform

(A) What is the ratio of the speed of the car in situations I, II and III?

- (a) 1 : 2 : 4 (b) 2 : 1 : 4
(c) 4 : 2 : 1 (d) 2 : 4 : 1

(C) What is the acceleration of the car in situation IV?

- (a) 1.667 m/s^2 (b) 16.67 m/s^2
(c) 1.667 cm/s^2 (d) 16.67 cm/s^2

Ans. (A) K

	I	II	III	IV
(b)	Uniform	Uniform	Uniform	Non-uniform

Explanation: Distance between two oil drops and consecutive drops is same in the first three situations so it means car is travelling at a uniform motion in I, II and III. As we see in situation IV, the distance between two oil drops and consecutive drops vary. Hence, in this situation, the car is not moving in uniform motion.

(B) (b) 2 : 1 : 4

Explanation: Speed of a car is the distance travelled per unit time.

$$\text{Speed of car} = \frac{\text{Distance between two consecutive points}}{\text{Time interval}}$$

In situation I,

Distance between two points is 2 which remains same.

(4 – 2 = 2, or 10 – 8 = 2 and so on)

∴ Distance travelled is 2 cm.

$$\text{Speed of car} = \frac{\text{Distance}}{\text{Time interval}} = \frac{2}{1} \\ = 2 \text{ cm/s}$$

In situation II,

Distance between two consecutive points = 1 cm

(2–1, 3–2 and so on)

∴ Distance travelled is 1 cm.

$$\text{So, the speed of car} = \frac{1}{1} = 1 \text{ cm/s}$$

In situation III,

Distance between two consecutive points = 4 cm

(8–4, 16–12 and so on)

∴ Distance travelled is 4 cm.

$$\text{So, the speed of car} = \frac{4}{1} = 4 \text{ cm/s}$$

Therefore, the ratio of speed of car in situations I, II and III is 2 cm/s, 1 cm/s, and 4 cm/s.

= 2 : 1 : 4

(B) (b) 16.67 m/s^2

Explanation: In situation D,

$$\text{Time between each drop} = \frac{1}{50} = 0.02 \text{ s}$$

Initial distance between two drops (u) = 2 cm or 0.02 m

$$\text{Speed} = \frac{\text{Distance}}{\text{Time}} = \frac{0.02}{0.02} = 1 \text{ m/s}$$

Final distance between two drops (v) = 4 cm or 0.04 m

$$\text{Speed} = \frac{0.04}{0.02} = 2 \text{ m/s}$$

$$\text{Time} = 6 \times 0.02 = 0.12 \text{ s}$$

$$v = u + at$$

$$a = \frac{v - u}{t} = \frac{2 - 1}{0.12} = \frac{1}{0.12} = 8.33 \text{ m/s}^2$$

VERY SHORT ANSWER Type Questions (VSA)

[1 mark each]

6. What happens to the speed of a bicycle when a brake is applied?

Ans. The application of brakes decelerates the bicycle and the speed is reduced.

Explanation: When the brakes are applied, the friction force between the cycle wheel and brake shoes decreases the speed of the wheel.



Related Theory

→ The rate at which the velocity changes is called acceleration or deceleration.

SHORT ANSWER Type-I Questions (SA-I)

[2 marks each]

7. For an object having uniform velocity, how will the first equation of motion change? [Diksha]

Ans. The first equation of motion is

$$v = u + at$$

For uniform motion, $a = 0$

$$v = u + (0) t$$

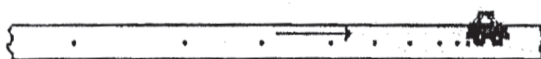
So, $v = u$

SHORT ANSWER Type-II Questions (SA-II)

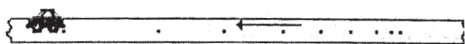
[3 marks each]

8. Draw the oil drop diagram of a car moving to the right and slowing down and a car moving to the left and speeding up. What is the difference between the two? [Delhi Gov. QB 2022]

Ans. (1) Moving right and slowing down



(2) Moving left and speeding up



These are the two ways to accelerate leftward: regardless of how it happens, the oil drop trace would be identical.

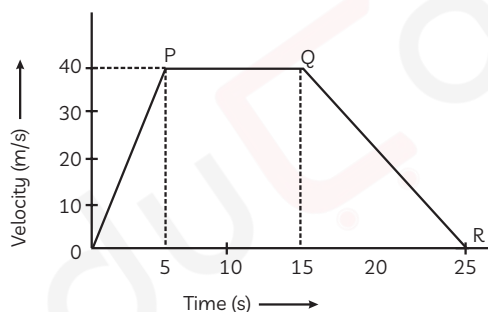
There is no difference.

The two diagrams would be indistinguishable from each other as indicated by the following diagrams.

LONG ANSWER Type Questions (LA)

[4 & 5 marks each]

9. In a given velocity-time graph of a moving bike, answer the following questions.



Velocity-Time graph

- (A) What type of motion is represented by OP?
- (B) What type of motion is represented by PQ?
- (C) What type of motion is represented by QR?
- (D) Find the acceleration of the bike from (i) O to P and (ii) Q to R.
- (E) Find the acceleration of the car from P to Q.

Ans. (A) The slope of the velocity-time graph represents the uniform acceleration.

(B) From the given graph, we can see that velocity remains unchanged with respect to time. So, PQ represents the constant velocity.

(C) From the given graph, we can see that velocity is decreasing. So, QR represents the uniform deceleration.

(D) From the given graph:

(i) Initial velocity, $u = 0 \text{ m/s}$

Final velocity, $v = 40 \text{ m/s}$

Total time, $t = 5 \text{ sec}$

$$\text{Acceleration} = \frac{\text{Change in velocity}}{\text{Time taken}}$$

$$a = \frac{v - u}{t}$$

$$\Rightarrow a = \frac{40 - 0}{5}$$

$$\Rightarrow a = 8 \text{ m/s}^2$$

The uniform acceleration of the bike is 8 m/s^2 .

(ii) Initial velocity, $u = 40 \text{ m/s}$

Final velocity, $v = 0 \text{ m/s}$

Total time, $t = (25 - 15) \text{ sec} = 10 \text{ sec}$

$$\text{Acceleration} = \frac{\text{Change in velocity}}{\text{Time taken}}$$

$$a = \frac{v - u}{t}$$

$$\Rightarrow a = \frac{0 - 40}{10}$$

$$\Rightarrow a = -4 \text{ m/s}^2$$

Uniform retardation of the bike is 4 m/s^2 .

(E) Initial velocity, $u = 40 \text{ m/s}$

Final velocity, $v = 40 \text{ m/s}$

Total time, $t = (15 - 5) \text{ sec}$
 $= 10 \text{ sec}$

$$\text{Acceleration} = \frac{\text{Change in velocity}}{\text{Time taken}}$$

$$a = \frac{v - u}{t}$$

$$\text{Acceleration} = \frac{\text{Change in velocity}}{\text{Time taken}}$$

$$a = \frac{40 - 40}{10}$$

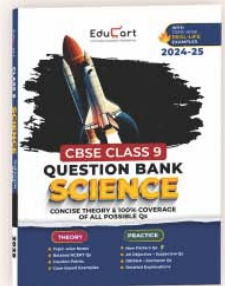
$$a = 0 \text{ m/s}^2$$

Acceleration is zero as a change in velocity is zero.



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